Chit 1

Chit 1

Use MySQL –(Joins and Subqueries)

Create Customer and Account table and add rows shown below

Customer Table

C_Id	Cname	City
1	John	Nashik
2	Seema	Aurangabad
3	Amita	Nagar
4	Rakesh	Pune
5	Samata	Nashik
6	Ankita	Chandwad
7	Bhavika	Pune
8	Deepa	Mumbai
9	Nitin	Nagpur
10	Pooja	Pune

Account Table

C_Id	Acc_Type	Amount
1	Current	5000
2	Saving	20000
3	Saving	70000
4	Saving	50000
6	Current	35000
7	Loan	30000
8	Saving	50000
9	Saving	90000
10	Loan	8000
11	Current	45000

1. Show the cname, Acc_Type, amount information of customer who is having an saving account.
2. Display the data using Natural, left and right join.
3. Display the information of customers living in the same city as of 'pooja'.
4. Display the information of account, having less amount than average amount throughout the bank.
5. Display the C_id having maximum amount in account.
6. Display the amount and acc_type of those customers whose amount is the minimum amount of that Acc_type.
7. Display the amount of those accounts whose amount is higher than amount of any saving account amount.

Create database

```
CREATE DATABASE chit1;
USE chit1;

CREATE TABLE Customer(
C_Id INT PRIMARY KEY,
Cname VARCHAR(20),
City VARCHAR(20)
);

CREATE TABLE Account(
C_Id INT,
Acc_Type VARCHAR(20),
Amount INT
);
```

Insert data

```
INSERT INTO Customer VALUES
(1,'John','Nashik'),
(2,'Seema','Aurangabad'),
(3,'Amita','Nagar'),
(4,'Rakesh','Pune'),
(5,'Samata','Nashik'),
(6,'Ankita','Chandwad'),
(7,'Bhavika','Pune'),
(8,'Deepa','Mumbai'),
(9,'Nitin','Nagpur'),
(10,'Pooja','Pune');
```

```
INSERT INTO Account VALUES
(1,'Current',5000),
(2,'Saving',20000),
(3,'Saving',70000),
(4,'Saving',5000),
(6,'Current',35000),
(7,'Loan',30000),
(8,'Saving',50000),
(9,'Saving',90000),
(10,'Loan',8000),
(11,'Current',45000);
```

1. Show the cname, Acc_Type, amount information of customer who is having an saving account

```
SELECT cname,Acc_Type,Amount
FROM Customer NATURAL JOIN Account
WHERE Acc_Type='Saving';
```

2. Display the data using Natural, left and right join.

```
SELECT * FROM Customer NATURAL JOIN Account;

SELECT * FROM Customer
LEFT JOIN Account
ON Customer.C_Id=Account.C_Id;

SELECT * FROM Customer
RIGHT JOIN Account
ON Customer.C_Id=Account.C_Id;
```

3. Display the information of customers living in the same city as of 'pooja'

```
SELECT * FROM Customer
WHERE City=(SELECT City FROM Customer WHERE Cname='Pooja');
```

4. Display the information of account, having less amount than average amount throughout the bank.

```
SELECT * FROM Account
WHERE Amount < (SELECT AVG(Amount) FROM Account);
```

5. Display the C_id having maximum amount in account.

```
SELECT C_Id FROM Account
WHERE Amount=(SELECT MAX(Amount) FROM Account);
```

6. Display the amount and acc_type of those customers whose amount is the minimum amount of that Acc_type.

```
SELECT Amount,Acc_Type
FROM Account A
WHERE Amount=(SELECT MIN(Amount)
FROM Account
WHERE Acc_Type=A.Acc_Type);
```

7. Display the amount of those accounts whose amount is higher than amount of any saving account amount.

```
SELECT Amount
FROM Account
WHERE Amount > (SELECT MIN(Amount) FROM Account);
```

Chit 4

Chit 4

(Perform on MYSQL Terminal)

student(S_ID,name,dept_name,tot_cred)

instructor(T_ID,name,dept_name,salary)

course(course_id,title,dept_name,credits)

i. Find the average salary of instructor in those departments where the average salary is more than Rs. 42000/-.
ii. Increase the salary of each instructor in the computer department by 10%.
iii. Find the names of instructors whose names are neither 'Amol' nor 'Amit'.
iv. Find the names of student which contains 'am' as its substring.
v. Find the name of students from computer department that "DBMS" courses they take.

Create database

```
CREATE DATABASE chit4;
USE chit4;

CREATE TABLE student(
S_ID INT,
name VARCHAR(20),
dept_name VARCHAR(20),
tot_cred INT
);

CREATE TABLE instructor(
T_ID INT,
name VARCHAR(20),
dept_name VARCHAR(20),
salary INT
);

CREATE TABLE course(
course_id INT,
title VARCHAR(20),
dept_name VARCHAR(20),
credits INT
);
```

Insert data

```
INSERT INTO instructor VALUES
(1,'Amol','Computer',50000),
(2,'Amit','IT',40000),
(3,'Raj','Computer',60000);

INSERT INTO student VALUES
(1,'Aman','Computer',100),
(2,'Ram','IT',90),
(3,'Sam','Computer',95);

INSERT INTO course VALUES
(101,'DBMS','Computer',5),
```

(102,'Java','IT',4);

i. Find the average salary of instructor in those departments where the average salary is more than Rs. 42000/-.

```
SELECT dept_name,AVG(salary)
FROM instructor
GROUP BY dept_name
HAVING AVG(salary)>42000;
```

ii. Increase the salary of each instructor in the computer department by 10%.

```
UPDATE instructor
SET salary=salary+salary*0.10
WHERE dept_name='Computer';
```

To display

```
SELECT * FROM instructor;
```

iii. Find the names of instructors whose names are neither ‘Amol’ nor ‘Amit’

```
SELECT name FROM instructor
WHERE name NOT IN ('Amol','Amit');
```

iv. Find the names of student which contains ‘am’ as its substring.

```
SELECT name FROM student
WHERE name LIKE '%am%';
```

v. Find the name of students from computer department that “DBMS” courses they take.

```
SELECT name FROM student
WHERE dept_name='Computer';
```

Chit 8

Chit 8

(Perform on MYSQL Terminal)

teaches(T_ID, course_id, sec_id, semester, year)

student(S_ID, name, dept_name, tot_cred)

instructor(T_ID, name, dept_name, salary)

course(course_id, title, dept_name, credits)

i. Find the names of the instructor in the university who have taught the courses semester wise.
ii. Create View on single table which retrieves student details.
iii. Rename the column of table student from dept_name to deptatrment_name
iv. Delete student name whose department is NULL

Create database
CREATE DATABASE chit8;
USE chit8;

CREATE TABLE teaches(
T_ID INT,
course_id INT,
sec_id INT,
semester VARCHAR(20),
year INT
);

CREATE TABLE student(
S_ID INT,
name VARCHAR(20),
dept_name VARCHAR(20),
tot_cred INT
);

CREATE TABLE instructor(
T_ID INT,
name VARCHAR(20),
dept_name VARCHAR(20),
salary INT
);

CREATE TABLE course(
course_id INT,
title VARCHAR(20),
dept_name VARCHAR(20),
credits INT
);

Insert data
INSERT INTO instructor VALUES
(1,'Amol','Computer',50000),

```
(2,'Amit','IT',45000),
(3,'Raj','Computer',60000);
INSERT INTO student VALUES
(101,'Prajwal','Computer',100),
(102,'Rahul','IT',90),
(103,'Aman',NULL,95);


INSERT INTO course VALUES
(201,'DBMS','Computer',5),
(202,'Java','IT',4);

INSERT INTO teaches VALUES
(1,201,1,'Winter',2025),
(2,202,2,'Summer',2025),
(3,201,1,'Winter',2025);
```

i. Find the names of the instructor in the university who have taught the courses semester wise

```
SELECT instructor.name, teaches.semester
FROM instructor
JOIN teaches
ON instructor.T_ID = teaches.T_ID;
```

ii. Create View on single table which retrieves student details

```
CREATE VIEW student_view AS
SELECT * FROM student;
```

To display the view:

```
SELECT * FROM student_view;
```

iii. Rename the column of table student from dept_name to deptatrment_name

```
ALTER TABLE student
CHANGE dept_name department_name VARCHAR(20);
```

iv. Delete student name whose department is NULL

```
DELETE FROM student
WHERE department_name IS NULL;
```

Chit16

Chit 16

Use MySQL Create Employee table, Project table and add rows shown below

Employee table

Eid	EName	Address	Salary	Commision
1	Amit	Pune	35000	5000
2	Sneha	Pune	25000	
3	Savita	Nasik	28000	2000
4	Pooja	Mumbai	19000	
5	Sagar	Mumbai	25000	3000

Project table

PrNo	Addr
10	Mumbai
20	Pune
30	Jalgoan

1. Find different locations from where employees belong to?
2. What is maximum and minimum salary?
3. Display the content of employee table according to the ascending order of salary amount.
4. Find the name of employee who lived in Nasik or Pune city.
5. Find the name of employees who does not get commission.
6. Change the city of Amit to Nashik.
7. Find the information of employees whose name starts with 'A'.
8. Find the count of staff from Mumbai.
9. Find the count of staff from each city
10. Find the address from where employees are belonging as well as where projects are going on.
11. Find city wise minimum salary.
12. Find city wise maximum salary having maximum salary greater than 26000
13. Delete the employee who is having salary greater than 30,000.

```
CREATE DATABASE chit16;
USE chit16;

CREATE TABLE Employee(
Eid INT,
Ename VARCHAR(20),
City VARCHAR(20),
Salary INT,
Commission INT
);
```

Insert Data

```
INSERT INTO Employee VALUES
(1,'Amit','Nashik',25000,NULL),
(2,'Rahul','Pune',30000,2000),
(3,'Sneha','Mumbai',28000,NULL),
(4,'Ajay','Pune',35000,1000);
```

1. Find different locations from where employees belong to?

```
SELECT DISTINCT City FROM Employee;
```

2. What is maximum and minimum salary?

```
SELECT MAX(Salary),MIN(Salary) FROM Employee;
```

3. Display the content of employee table according to the ascending order of salary

```
SELECT * FROM Employee
ORDER BY Salary ASC;
```

4. Find the name of employee who lived in Nasik or Pune city

```
SELECT Ename FROM Employee
WHERE City IN ('Nashik','Pune');
```

5. Find the name of employees who does not get commission

```
SELECT Ename FROM Employee
WHERE Commission IS NULL;
```

6. Change the city of Amit to Nashik

```
UPDATE Employee
SET City='Nashik'
WHERE Ename='Amit';
```

7. Find the information of employees whose name starts with 'A'.

```
SELECT * FROM Employee
WHERE Ename LIKE 'A%';
```

8. Find the count of staff from Mumbai.

```
SELECT COUNT(*) FROM Employee
WHERE City='Mumbai';
```

9. Find the count of staff from each city

```
SELECT City,COUNT(*) FROM Employee
GROUP BY City;
```

11. Find city wise minimum salary

```
SELECT City,MIN(Salary)
FROM Employee
GROUP BY City;
```

12. Find city wise maximum salary having maximum salary greater than 26000

```
SELECT City,MAX(Salary)
FROM Employee
GROUP BY City
HAVING MAX(Salary)>26000;
```

13. Delete the employee who is having salary greater than 30,000.

```
DELETE FROM Employee
WHERE Salary>30000;
```

Chit 17

Chit 17

Use MySQL

Create a table emp with following fields and constraints

Eno –(Constraint:- primary key and apply sequence starts with 101) ,Ename –(Constraint :- not null)

Address —(Constraint :-default 'Nashik') ,Joindate,

After table creation add field - Post in the emp table.

Insert some data in emp table.Create Index on Ename field of employee table.

Create View on employee table to show only Ename and Salary.

```
CREATE DATABASE chit17;
USE chit17;

CREATE TABLE emp(
Eno INT AUTO_INCREMENT PRIMARY KEY,
Ename VARCHAR(20) NOT NULL,
Address VARCHAR(20) DEFAULT 'Nashik',
Joindate DATE,
Salary INT
) AUTO_INCREMENT = 101;
```

After table creation add field - Post in the emp table.

```
ALTER TABLE emp
ADD Post VARCHAR(20);
```

Insert some data in emp table

```
INSERT INTO emp(Ename, Joindate, Salary, Post)
VALUES
('Amit','2024-01-10',25000,'Manager'),
('Rahul','2024-02-15',30000,'Clerk');
```

.Create Index on Ename field of employee table

```
CREATE INDEX idx_ename
ON emp(Ename);
```

Create View on employee table to show only Ename and Salary

```
CREATE VIEW emp_view AS
SELECT Ename, Salary
FROM emp;

SELECT * FROM emp_view;
```

Chit 19

Chit 19:

(Perform on MYSQL Terminal)

Emp(emp_id,ename, street, city)

works(emp_id,c_id,ename, cname, sal)

Company(c_id,cname, city)

Manager(mgr_id, mgrname)

i. Modify the database so that a particular company (eg. ABC) now in Pune
ii. Give all managers of Mbank a 10% raise. If salary is >20,000, give only 3% raise.
iii. Find out the names of all the employees who works in 'Bosch' company in city Pune
iv. Delete all records in the works table for employees of a particular company (Eg, SBC Company) whose salary>50,000.

```
CREATE DATABASE chit19;
USE chit19;

CREATE TABLE Emp(
emp_id INT,
ename VARCHAR(20),
street VARCHAR(20),
city VARCHAR(20)
);


CREATE TABLE works(
emp_id INT,
c_id INT,
ename VARCHAR(20),
cname VARCHAR(20),
sal INT
);


CREATE TABLE Manager(
mgr_id INT,
mgrname VARCHAR(20)
);


CREATE TABLE Company(
c_id INT,
cname VARCHAR(20),
city VARCHAR(20)
);

INSERT INTO Emp VALUES
(1,'Amit','MG Road','Pune'),
(2,'Rahul','College Road','Mumbai'),
(3,'Prajwal','Station Road','Pune');
```

```
INSERT INTO Company VALUES
(101,'ABC','Mumbai'),
(102,'Bosch','Pune'),
(103,'SBC','Delhi'),
(104,'Mbank','Mumbai');
```

```
INSERT INTO works VALUES
(1,102,'Amit','Bosch',40000),
(2,103,'Rahul','SBC',60000),
(3,104,'Prajwal','Mbank',18000),
(4,104,'Rohit','Mbank',25000);
```

i. Modify the database so that a particular company (eg. ABC) now in Pune

```
UPDATE Company
SET city='Pune'
WHERE cname='ABC';
```

10% raise if salary <= 20000

```
UPDATE works
SET sal = sal + sal*0.10
WHERE cname='Mbank'
AND sal <= 20000;
```

3% raise if salary > 20000

```
UPDATE works
SET sal = sal + sal*0.03
WHERE cname='Mbank'
AND sal > 20000;
```

iii. Find out the names of all the employees who works in ‘Bosch’ company in city Pune

```
SELECT ename
FROM works
WHERE cname='Bosch';
```

iv. Delete all records in the works table for employees of a particular company (Eg, SBC Company) whose salary>50,000.

```
DELETE FROM works
WHERE cname='SBC'
AND sal > 50000;
```

Chit 21

Chit 21:

(Perform on MYSQL Terminal)

Empl(e_no, e_name, post, pay_rate)

Position(pos_no, post)

Duty-alloc (pos_no, e_no, month,year, shift)

Implement the following SQL queries

i. Get duty allocation details for e_no 123 for the first shift in the month of April 2003

ii. Get the employees whose rate of pay is > or equal rate of pay of employees 'Sachin'.

iii. Create a view for displaying minimum, maximum and average salary for all the posts.

iv. Get count of different employees on each shift having post 'manager'.

Create Database

```
CREATE DATABASE chit21;
USE chit21;

CREATE TABLE Empl(
e_no INT,
e_name VARCHAR(20),
post VARCHAR(20),
pay_rate INT
);

CREATE TABLE Positions(
pos_no INT,
post VARCHAR(20)
);

CREATE TABLE Duty_alloc(
pos_no INT,
e_no INT,
month VARCHAR(20),
year INT,
shift INT
);
```

Insert Data

```
INSERT INTO Empl VALUES
(123,'Sachin','Manager',50000),
(124,'Rahul','Clerk',30000),
(125,'Amit','Manager',55000);

INSERT INTO Positions VALUES
(1,'Manager'),
(2,'Clerk');

INSERT INTO Duty_alloc VALUES
(1,123,'April',2003,1),
(2,124,'May',2003,2),
(1,125,'April',2003,1);
```

i. Get duty allocation details for e_no 123 for the first shift in the month of April 2003

```
SELECT * FROM Duty_alloc
WHERE e_no=123
AND shift=1
AND month='April'
AND year=2003;
```

ii. Get the employees whose rate of pay is > or equal rate of pay of employees 'Sachin'.

```
SELECT * FROM Empl
WHERE pay_rate >=
(SELECT pay_rate FROM Empl
WHERE e_name='Sachin');
```

iii. Create a view for displaying minimum, maximum and average salary for all the posts

```
CREATE VIEW salary_view AS
SELECT post,
MIN(pay_rate) AS Minimum_Salary,
MAX(pay_rate) AS Maximum_Salary,
AVG(pay_rate) AS Average_Salary
FROM Empl
GROUP BY post;
```

Display View

```
SELECT * FROM salary_view;
```

iv. Get count of different employees on each shift having post ‘manager’

```
SELECT shift, COUNT(DISTINCT D.e_no)
FROM Duty_alloc D, Empl E
WHERE D.e_no = E.e_no
AND E.post = 'Manager'
GROUP BY shift;
```